

GARP Risk & AI — Supplementary Memory & Theory Questions

Topics Not Covered in the Main 150-Question Deck

Unofficial

October 3, 2026

Question

The term “artificial intelligence” was coined in 1956 by:

- A. Alan Turing
- B. John McCarthy
- C. Marvin Minsky
- D. Claude Shannon

Answer: B

Explanation

John McCarthy of Dartmouth College coined the term in 1956, along with Marvin Minsky, Nathan Rochester, and Claude Shannon, at the seminal Dartmouth Summer Research Project.

Question

The Logic Theorist program, designed to generate proofs in propositional logic, was developed by:

- A. Newell, Simon, and Shaw
- B. McCarthy, Minsky, and Shannon
- C. McCulloch and Pitts
- D. Rosenblatt

Answer: A

Explanation

Allen Newell, Herbert Simon, and Cliff Shaw developed Logic Theorist at the Rand Corporation in 1955. It confirmed most theorems in Chapter 2 of Principia Mathematica.

Question

The 1943 paper “A logical calculus of the ideas immanent in nervous activity” was authored by:

- A. McCulloch and Pitts
- B. Hebb
- C. Rosenblatt
- D. Minsky and Papert

Answer: A

Explanation

Warren McCulloch and Walter Pitts published this foundational paper in the Bulletin of Mathematical Biology, describing neurons and neural networks in terms of propositional logic.

Question

Donald Hebb's 1949 book that proposed associative learning in neural networks is titled:

- A. The Organisation of Behaviour
- B. Perceptrons
- C. Parallel Distributed Processing
- D. Computing Machinery and Intelligence

Answer: A

Explanation

Hebb's "The Organisation of Behaviour" (1949) proposed the neurophysiological postulate that when one neuron repeatedly fires another, the connection strengthens.

Question

Frank Rosenblatt's Perceptron was formulated in terms of:

- A. Symbolic logic
- B. Probability theory
- C. Propositional calculus
- D. Predicate logic

Answer: B

Explanation

Rosenblatt (1958) formulated the Perceptron "in terms of probability theory rather than symbolic logic," rejecting symbolic or algorithmic approaches.

Question

Minsky and Papert's 1969 book that severely undermined enthusiasm for neural networks was titled:

- A. Perceptrons
- B. The Organisation of Behaviour
- C. Parallel Distributed Processing
- D. Superintelligence

Answer: A

Explanation

Minsky and Papert's "Perceptrons" (1969) showed that single-layer Perceptrons could not learn simple logical functions like XOR.

Question

The 1986 two-volume work that displaced pessimism about neural networks and encouraged connectionism was:

- A. Parallel Distributed Processing
- B. Perceptrons
- C. The Organisation of Behaviour
- D. Computing Machinery and Intelligence

Answer: A

Explanation

Rumelhart, McClelland, and the PDP Research Group published "Parallel Distributed Processing" in 1986, demonstrating how layers of interacting neurons could learn complex functions.

Question

LeNet-5, a convolutional neural network designed for recognition of digits, was developed by:

- A. Yann LeCun and colleagues
- B. Krizhevsky, Sutskever, and Hinton
- C. DeepMind
- D. OpenAI

Answer: A

Explanation

Yann LeCun and colleagues published LeNet-5 in 1998, consisting of 8 layers with convolutional and pooling layers.

Question

AlexNet won the 2012 ImageNet Large Scale Visual Recognition Challenge with an error rate of:

- A. 25.8%
- B. 15.3%
- C. 10.0%
- D. 5.0%

Answer: B

Explanation

AlexNet (Krizhevsky, Sutskever, Hinton) achieved a 15.3% error rate in 2012, compared to the previous year's winning 25.8%.

Question

In 2016, AlphaGo defeated World Go Champion Lee Sedol with a score of:

- A. 5-0
- B. 4-1
- C. 3-2
- D. 2-3

Answer: B

Explanation

AlphaGo defeated Lee Sedol 4-1 in 2016. It had previously defeated European champion Fan Hui 5-0 in 2015.

Question

AlphaZero learned to play chess, Go, and Shogi by:

- A. Learning from human game databases
- B. Competing against itself
- C. Following expert-designed heuristics
- D. Using exhaustive search

Answer: B

Explanation

AlphaZero learned by self-play (competing against itself) without any input from human experts or game databases, defeating Stockfish after only a few hours.

Question

The term “stochastic parrot” was coined by:

- A. Emily M. Bender
- B. Geoffrey Hinton
- C. Yann LeCun
- D. Fei Fei Li

Answer: A

Explanation

Emily M. Bender coined the term “stochastic parrot” in 2021 to describe LLMs that imitate language without understanding meaning.

Question

Alan Turing's 1950 paper "Computing Machinery and Intelligence" introduced the Turing Test under what original name?

- A. The Imitation Game
- B. The Reasoning Test
- C. The Intelligence Game
- D. The Machine Test

Answer: A

Explanation

Turing called it the "imitation game," where a computer proves intelligence if its textual conversation is indistinguishable from a human's.

Question

Alan Turing's 1936 paper introducing the "Turing machine" was titled:

- A. On Computable Numbers, with an Application to the Entscheidungsproblem
- B. Computing Machinery and Intelligence
- C. The Organisation of Behaviour
- D. A Logical Calculus of the Ideas Immanent in Nervous Activity

Answer: A

Explanation

Turing's seminal 1936 paper appeared in the Proceedings of the London Mathematical Society and introduced the theoretical computing machine.

Question

The K-means algorithm is also known as:

- A. Lloyd's algorithm
- B. Newton's algorithm
- C. Bayes' algorithm
- D. Fisher's algorithm

Answer: A

Explanation

K-means is also known as Lloyd's algorithm, after Stuart Lloyd who developed it at Bell Labs in 1957.

Question

A rough guideline for initially estimating K in K-means is:

- A. $K = N/2$
- B. $K = \sqrt{N/2}$
- C. $K = \log(N)$
- D. $K = 2N$

Answer: B

Explanation

A rough guideline is $K = \sqrt{N/2}$, where N is the number of data points. This is arbitrary and more sophisticated methods are preferred.

Question

K-means++ is an improved initialization method whose purpose is to:

- A. Reduce K
- B. Spread initial centroids far apart to reduce poor local optima
- C. Increase WCSS
- D. Eliminate outliers

Answer: B

Explanation

K-means++ selects initial centroids with probability proportional to squared distance from nearest existing centroid, spreading them out and reducing poor local optima.

Question

BCSS measures:

- A. Intra-cluster compactness
- B. Sum of squared distances of each cluster's centroid to the overall centroid, weighted by cluster size
- C. Total variance
- D. Number of clusters

Answer: B

Explanation

BCSS focuses on inter-cluster dissimilarity (separation). A higher BCSS suggests clusters are well separated.

Question

Single linkage defines distance between two clusters as:

- A. The greatest distance between any two points in different clusters
- B. The shortest distance between any two points in different clusters
- C. The average distance between all points
- D. The distance between centroids

Answer: B

Explanation

Single linkage = minimum distance between clusters. Complete linkage = maximum distance. Average linkage = average distance.

Question

Complete linkage defines distance between two clusters as:

- A. The shortest distance between any two points
- B. The greatest distance between any two points in different clusters
- C. The average distance
- D. The distance between medoids

Answer: B

Explanation

Complete linkage uses the maximum distance between clusters. It tends to produce compact, spherical clusters.

Question

Agglomerative clustering starts with:

- A. All data in one cluster and splits
- B. Each data point as its own cluster and merges
- C. Random clusters
- D. K clusters

Answer: B

Explanation

Agglomerative = bottom-up (each point starts as its own cluster, then merge). Divisive = top-down (start with one cluster, then split).

Question

In DBSCAN, a core point is defined as:

- A. Any point in the dataset
- B. A point with at least a minimum number of other points within a specified radius
- C. A point on the cluster boundary
- D. An outlier

Answer: B

Explanation

A core point has at least `minPts` neighbors within radius `epsilon`. Border points are within `epsilon` of a core point. Noise points are neither.

Question

In DBSCAN, a noise point is:

- A. A core point
- B. A border point
- C. A point that is neither a core point nor a border point
- D. A centroid

Answer: C

Explanation

Noise points (outliers) are in low-density regions and are not assigned to any cluster in DBSCAN.

Question

The Gini coefficient for a node with J classes is:

- A. $1 - \sum_{j=1}^J p_j^2$
- B. $-\sum_{j=1}^J p_j \log_2(p_j)$
- C. $\sum_{j=1}^J p_j$
- D. $\max_j p_j$

Answer: A

Explanation

Gini = 1 minus sum of squared proportions. Entropy = negative sum of $p \log p$. Both measure node impurity in classification trees.

Question

Entropy for a node with J classes is:

- A. $1 - \sum p_j^2$
- B. $-\sum_{j=1}^J p_j \log_2(p_j)$
- C. $\sum p_j$
- D. $\max p_j$

Answer: B

Explanation

Entropy measures disorder. Base-2 log is used; changing base scales all values by a constant.

Question

Pre-pruning (online pruning) in decision trees:

- A. Grows the tree fully then removes branches
- B. Stops tree growth early using stopping criteria
- C. Removes outliers
- D. Increases tree depth

Answer: B

Explanation

Pre-pruning prevents a tree from growing too much (e.g., minimum observations per node). Post-pruning grows fully then removes weak branches.

Question

Reduced error pruning is a:

- A. Top-down approach
- B. Bottom-up approach that replaces nodes with most popular class if validation performance doesn't worsen
- C. Type of boosting
- D. Type of bagging

Answer: B

Explanation

Reduced error pruning is bottom-up: start at leaves, replace nodes with their most popular class if the pruned tree performs no worse on validation data.

Question

Cost complexity pruning adds a penalty term to RSS, with the trade-off controlled by:

- A. alpha
- B. lambda
- C. eta
- D. mu

Answer: A

Explanation

Cost complexity pruning uses parameter alpha, chosen by cross-validation, to trade off accuracy vs number of terminal nodes.

Question

In a random forest, the number of features considered at each split is typically:

- A. All features
- B. Approximately the square root of the total number of features
- C. One feature
- D. Half the features

Answer: B

Explanation

Random forest uses a random subset of features at each split, usually approximately the square root of total features. This reduces correlation across trees.

Question

Pasting differs from bagging because it:

- A. Samples with replacement
- B. Samples without replacement
- C. Uses all data
- D. Uses no data

Answer: B

Explanation

Bagging = sampling with replacement. Pasting = sampling without replacement. Both create multiple subsamples for ensemble training.

Question

KNN is called a “lazy learner” because:

- A. It trains very slowly
- B. It doesn't build an explicit model during training; computation happens at prediction time
- C. It uses few features
- D. It ignores training data

Answer: B

Explanation

KNN memorizes the training dataset and performs all computation at prediction time when distances are calculated.

Question

A common choice for K in KNN is:

- A. $K = N$
- B. $K = \text{sqrt}(N)$
- C. $K = 1$
- D. $K = N/2$

Answer: B

Explanation

$K = \text{sqrt}(N)$ is a common heuristic. Small K overfits; large K underfits. Cross-validation can tune K .

Question

Support vectors are:

- A. All training points
- B. The training points closest to the decision boundary that define the margin
- C. The centroids of classes
- D. Outliers

Answer: B

Explanation

Support vectors lie on the margin edges and determine the position and orientation of the decision boundary.

Question

The maximum margin classifier is:

- A. Any separating line
- B. The line equidistant from margin constraints that maximizes the margin
- C. The line through the origin
- D. The line minimizing margin

Answer: B

Explanation

The maximum margin classifier is the optimal decision boundary that maximizes the margin between classes.

Question

In SVM soft margin, a large lambda leads to:

- A. Narrower margin, fewer misclassifications
- B. Wider margin, more misclassifications allowed
- C. No margin
- D. Perfect separation

Answer: B

Explanation

Large lambda allows wider margin with more misclassifications (more tolerant). Small lambda penalizes misclassification heavily (narrower margin).

Question

The kernel trick in SVM allows:

- A. Linear classification only
- B. Nonlinear decision boundaries by projecting data to higher-dimensional space without explicit computation
- C. Removing outliers
- D. Reducing features

Answer: B

Explanation

The kernel trick computes inner products in higher-dimensional space without explicitly transforming data, enabling nonlinear boundaries.

Question

The universal approximation theorem states that:

- A. Neural networks can only approximate linear functions
- B. Neural networks with enough hidden neurons can approximate any function with arbitrary precision
- C. Neural networks cannot learn nonlinear functions
- D. Neural networks always overfit

Answer: B

Explanation

The theorem states that with the right number of hidden neurons, neural networks can approximate any function with arbitrary precision under some assumptions.

Question

The sigmoid (logistic) activation function is:

- A. $f(z) = \max(0, z)$
- B. $f(z) = 1/(1 + e^{-z})$
- C. $f(z) = (e^{2z} - 1)/(e^{2z} + 1)$
- D. $f(z) = z$

Answer: B

Explanation

Sigmoid outputs between 0 and 1. Used in binary classification output layers.

Question

The ReLU activation function is:

- A. $f(z) = 1/(1 + e^{-z})$
- B. $f(z) = \max(0, z)$
- C. $f(z) = \tanh(z)$
- D. $f(z) = z^2$

Answer: B

Explanation

ReLU = $\max(0, z)$. It is computationally efficient and less prone to vanishing gradient for positive values.

Question

The leaky ReLU activation function is:

- A. $f(z) = \max(0, z)$
- B. $f(z) = 0.01z$ if $z < 0$, z if $z \geq 0$
- C. $f(z) = 1/(1 + e^{-z})$
- D. $f(z) = \tanh(z)$

Answer: B

Explanation

Leaky ReLU allows small negative values ($0.01z$), preventing dead neurons that can occur with standard ReLU.

Question

The hyperbolic tangent ($\tanh$) activation function outputs values between:

- A. 0 and 1
- B. -1 and 1
- C. 0 and infinity
- D. -infinity and infinity

Answer: B

Explanation

Tanh outputs between -1 and 1, similar in shape to sigmoid but zero-centered.

Question

Dropout regularization works by:

- A. Removing features
- B. Randomly deactivating a fraction of neurons during training
- C. Increasing learning rate
- D. Adding more layers

Answer: B

Explanation

Dropout randomly drops neurons during training, preventing co-adaptation and reducing overfitting. At inference, all neurons are used.

Question

Early stopping in neural network training:

- A. Stops training when validation error starts to rise
- B. Stops after a fixed number of epochs
- C. Never stops
- D. Stops immediately

Answer: A

Explanation

Early stopping monitors validation performance and halts training when it starts to deteriorate, preventing overfitting.

Question

Batch normalization:

- A. Normalizes weights
- B. Adds normalization layers between hidden layers to stabilize training
- C. Removes features
- D. Increases learning rate

Answer: B

Explanation

Batch normalization normalizes inputs to each layer across the batch, stabilizing training and mitigating internal covariate shift.

Question

A convolutional layer in a CNN:

- A. Connects every neuron to every other neuron
- B. Applies a filter/kernel to local regions of the input
- C. Only works with 1D data
- D. Removes features

Answer: B

Explanation

Convolutional layers use filters that slide across the input, creating feature maps. This reduces parameters and builds translation invariance.

Question

A pooling layer in a CNN:

- A. Increases dimensions
- B. Reduces dimensions by summarizing regions (e.g., average or max)
- C. Adds parameters
- D. Removes activation functions

Answer: B

Explanation

Pooling layers reduce spatial dimensions by taking summary statistics (e.g., max, average) over regions, reducing computation.

Question

Autoencoders are primarily used for:

- A. Classification
- B. Dimensionality reduction and anomaly detection
- C. Regression
- D. Clustering only

Answer: B

Explanation

Autoencoders learn compressed representations (dimensionality reduction) and can detect anomalies via high reconstruction error.

Question

Autoencoders differ from PCA because they:

- A. Only perform linear transformations
- B. Can capture nonlinear relationships using activation functions
- C. Require no training
- D. Always produce orthogonal components

Answer: B

Explanation

Autoencoders with nonlinear activation functions can capture nonlinear patterns, unlike PCA which is strictly linear.

Question

The autoencoder loss function is:

- A. $\sum \sum (x_{ij} - \hat{x}_{ij})^2$
- B. $\sum |x_{ij} - \hat{x}_{ij}|$
- C. $\max(x_{ij} - \hat{x}_{ij})$
- D. $\sum x_{ij}$

Answer: A

Explanation

Autoencoder loss = reconstruction error. The goal is to minimize this error.

Question

OLS estimates parameters by minimizing:

- A. Sum of residuals
- B. Residual Sum of Squares (RSS)
- C. Mean Absolute Error
- D. Maximum likelihood

Answer: B

Explanation

OLS minimizes RSS. Squaring prevents positive and negative residuals from canceling.

Question

The OLS slope estimate is:

- A. $\hat{\beta}_1 = \bar{y} - \hat{\beta}_0 \bar{x}$
- B. $\hat{\beta}_1 = \frac{\sum y_i x_i - N \bar{y} \bar{x}}{\sum x_i^2 - N \bar{x}^2}$
- C. $\hat{\beta}_1 = \bar{y} / \bar{x}$
- D. $\hat{\beta}_1 = \sum (y_i - \bar{y})(x_i - \bar{x})$

Answer: B

Explanation

OLS slope = covariance(x,y)/variance(x). Option A is the intercept formula. Option D is missing the denominator.

Question

Given $\hat{\beta}_1$, the OLS intercept is:

- A. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- B. $\hat{\beta}_0 = \bar{y} + \hat{\beta}_1 \bar{x}$
- C. $\hat{\beta}_0 = \bar{y} / \hat{\beta}_1 \bar{x}$
- D. $\hat{\beta}_0 = \hat{\beta}_1 \bar{x}$

Answer: A

Explanation

The regression line passes through the point of means, so the intercept is the mean of y minus slope times mean of x.

Question

MLE estimates parameters by:

- A. Minimizing RSS
- B. Maximizing the likelihood of observing the data
- C. Minimizing absolute errors
- D. Maximizing R-squared

Answer: B

Explanation

MLE finds parameter values that maximize the likelihood function. For linear regression with normal errors, $\text{MLE} = \text{OLS}$.

Question

The log-likelihood for the logit model is:

- A. $\sum [y_i \ln F(y_i) + (1 - y_i) \ln(1 - F(y_i))]$
- B. $\sum y_i$
- C. $\sum (y_i - F(y_i))^2$
- D. $\prod F(y_i)$

Answer: A

Explanation

Log-likelihood sums log-probabilities. Option D is the likelihood before log. Option C is squared-error, not likelihood.

Question

The gradient descent weight update is:

- A. $w_i^{\text{new}} = w_i^{\text{old}} + \eta \frac{\partial L}{\partial w_i}$
- B. $w_i^{\text{new}} = w_i^{\text{old}} - \eta \frac{\partial L}{\partial w_i}$
- C. $w_i^{\text{new}} = w_i^{\text{old}} \cdot \eta$
- D. $w_i^{\text{new}} = \eta$

Answer: B

Explanation

Gradient descent subtracts the learning-rate-scaled gradient. Option A would be gradient ascent.

Question

Exponential decay of learning rate is:

- A. $\eta_t = \eta_0 + kt$
- B. $\eta_t = \eta_0 e^{-kt}$
- C. $\eta_t = \eta_0 \cdot kt$
- D. $\eta_t = \eta_0 / k$

Answer: B

Explanation

Exponential decay reduces learning rate over epochs. Inverse decay is another option.

Question

Inverse decay of learning rate is:

- A. $\eta_t = \eta_0 e^{-kt}$
- B. $\eta_t = \eta_0 / (1 + kt)$
- C. $\eta_t = \eta_0 (1 + kt)$
- D. $\eta_t = \eta_0 \cdot k$

Answer: B

Explanation

Inverse decay. Both exponential and inverse decay reduce learning rate over epochs.

Question

Backpropagation is used to:

- A. Forward propagate inputs
- B. Compute gradients efficiently using chain rule from output to input layers
- C. Cluster data
- D. Reduce features

Answer: B

Explanation

Backpropagation computes gradients layer-by-layer using the chain rule, starting from the output layer and working backward.

Question

The vanishing gradient problem occurs when:

- A. Gradients grow too large
- B. Gradients become very small as they propagate through many layers
- C. Weights are too large
- D. Learning rate is too high

Answer: B

Explanation

Vanishing gradient: gradients shrink exponentially through deep networks with saturating activations (e.g., sigmoid, tanh).

Question

The exploding gradient problem occurs when:

- A. Gradients shrink to zero
- B. Gradients grow extremely large across layers
- C. Weights are zero
- D. Learning rate is zero

Answer: B

Explanation

Exploding gradient: gradients grow exponentially. Remedies include gradient clipping and careful initialization.

Question

Momentum in gradient descent:

- A. Reduces learning rate
- B. Accelerates convergence in consistent directions and dampens oscillation
- C. Eliminates local minima
- D. Converts non-convex to convex

Answer: B

Explanation

Momentum accumulates velocity, accelerating consistent gradient directions and reducing oscillation.

Question

Adam optimizer combines:

- A. Momentum and adaptive learning rates
- B. Batch and stochastic descent
- C. Ridge and LASSO
- D. Precision and recall

Answer: A

Explanation

Adam = Adaptive Moment Estimation. It combines momentum (first moment) with adaptive per-parameter learning rates (second moment).

Question

An epoch in neural network training is:

- A. One gradient update
- B. One full pass through the entire training dataset
- C. One validation step
- D. One feature selection step

Answer: B

Explanation

An epoch = one full pass through the training data. Typically many epochs are required for convergence.

Question

Batch gradient descent updates weights using:

- A. One random training example
- B. A subset of training examples
- C. The entire training set
- D. No training examples

Answer: C

Explanation

Batch GD uses the full dataset per update. Stochastic uses one example. Mini-batch uses a subset.

Question

Stochastic gradient descent (SGD) updates weights using:

- A. The full training set
- B. One random training example per step
- C. A fixed subset
- D. No examples

Answer: B

Explanation

SGD uses one randomly selected example per update. It is noisier but faster per pass.

Question

Mini-batch gradient descent updates weights using:

- A. One example
- B. The full dataset
- C. A small subset of training data per step
- D. No data

Answer: C

Explanation

Mini-batch GD uses a small subset (e.g., 32 or 64 examples). It balances parallelism and stochasticity.

Question

Overfitting occurs when:

- A. The model is too simple
- B. The model captures noise in training data and fails to generalize
- C. The model has too few parameters
- D. The model underfits

Answer: B

Explanation

Overfitting = model captures noise, performs well on training but poorly on test data. Sign: large gap between training and test error.

Question

Underfitting occurs when:

- A. The model is too complex
- B. The model fails to capture relevant patterns in the data
- C. The model has too many parameters
- D. The model overfits

Answer: B

Explanation

Underfitting = model too simple to capture underlying patterns. High bias, low variance.

Question

The bias-variance trade-off states that:

- A. Increasing model complexity always reduces total error
- B. Increasing complexity reduces bias but increases variance
- C. Bias and variance are independent
- D. Variance is always zero

Answer: B

Explanation

Total error = Bias squared + Variance + Irreducible Error. More complex models have lower bias but higher variance.

Question

Cross-validation is used to:

- A. Increase training data
- B. Assess model generalization and tune hyperparameters
- C. Remove outliers
- D. Reduce features

Answer: B

Explanation

Cross-validation splits data into k folds, training on k minus 1 and validating on the held-out fold, to estimate out-of-sample performance.

Question

In k-fold cross-validation, the data is:

- A. Split into k equal folds; model trained k times, each time holding out a different fold
- B. Split into 2 folds only
- C. Not split
- D. Split randomly without reuse

Answer: A

Explanation

K-fold CV partitions into k folds; iterate k times, each time training on k-1 folds and validating on the held-out fold.

Question

Stratified cross-validation ensures:

- A. Each fold has the same class proportion as the whole dataset
- B. Only the majority class is used
- C. All folds have equal mean
- D. Random assignment is avoided

Answer: A

Explanation

Stratified CV preserves class ratios in every fold, critical for imbalanced datasets.

Question

Bootstrapping in model evaluation is:

- A. Sampling with replacement to create new datasets
- B. Sampling without replacement
- C. Splitting data into two halves
- D. Training without validation

Answer: A

Explanation

Bootstrapping draws samples with replacement. About 37 percent of observations are not sampled (out-of-bootstrap).

Question

In bootstrapping, the expected fraction of observations NOT sampled is approximately:

- A. 10 percent
- B. 25 percent
- C. 37 percent
- D. 63 percent

Answer: C

Explanation

Expected out-of-bootstrap fraction = $1 - 1/e = 0.368 = 37$ percent.

Question

Grid search is used for:

- A. Exhaustively searching through predefined hyperparameter combinations
- B. Removing outliers
- C. Feature selection only
- D. Clustering

Answer: A

Explanation

Grid search evaluates every combination of hyperparameters within specified ranges, often combined with cross-validation.

Question

AIC is defined as:

- A. $AIC = 2k - 2 \ln(L)$
- B. $AIC = k + L$
- C. $AIC = 2k + 2 \ln(L)$
- D. $AIC = L/k$

Answer: A

Explanation

$AIC = 2k - 2 \ln(L)$, where k = number of parameters, L = likelihood. Lower AIC = better model.

Question

BIC is defined as:

- A. $BIC = k \ln(N) - 2 \ln(L)$
- B. $BIC = 2k - 2 \ln(L)$
- C. $BIC = k + L$
- D. $BIC = L^2$

Answer: A

Explanation

$BIC = k \ln(N) - 2 \ln(L)$. It penalizes complexity more heavily than AIC for large samples.

Question

Forward stepwise selection:

- A. Starts with all features and removes them one by one
- B. Starts with no features and adds them one by one
- C. Uses all features always
- D. Randomly selects features

Answer: B

Explanation

Forward selection starts with no features and adds the feature that most improves the criterion (e.g., AIC). Backward elimination starts with all features.

Question

Backward stepwise elimination:

- A. Starts with no features and adds them
- B. Starts with all features and removes them one by one
- C. Uses no features
- D. Randomly selects features

Answer: B

Explanation

Backward elimination starts with the full model and removes the feature whose removal most improves the criterion.

Question

Tokenization in NLP is:

- A. Removing stop words
- B. Separating text into sentences and words
- C. Converting words to vectors
- D. Replacing words with synonyms

Answer: B

Explanation

Tokenization separates documents into sentences (sentence tokenization) and sentences into words (word tokenization).

Question

Stop word removal involves:

- A. Removing words with no informational value (e.g., the, a, and)
- B. Converting words to vectors
- C. Separating sentences
- D. Replacing words with roots

Answer: A

Explanation

Stop words are common words with little informational value removed to improve analysis efficiency.

Question

Stemming in NLP:

- A. Replaces words with their dictionary form
- B. Replaces words with their root form by cutting prefixes and suffixes
- C. Removes stop words
- D. Converts to vectors

Answer: B

Explanation

Stemming cuts prefixes/suffixes (e.g., disappointing becomes disappoint). Lemmatization uses dictionary forms.

Question

Lemmatization in NLP:

- A. Cuts prefixes and suffixes
- B. Replaces words with their dictionary form based on meaning and part of speech
- C. Removes stop words
- D. Converts to vectors

Answer: B

Explanation

Lemmatization uses vocabulary and morphological analysis to find the lemma (dictionary form) of a word.

Question

In binary bag of words, each element of the document vector is:

- A. The count of word occurrences
- B. 1 if the word appears, 0 otherwise
- C. The TF-IDF score
- D. The word embedding

Answer: B

Explanation

Binary BoW: 1 = word present, 0 = absent. Count BoW records the number of occurrences.

Question

In count bag of words, each element of the document vector is:

- A. 1 or 0
- B. The number of times the word appears in the document
- C. The TF-IDF score
- D. The word embedding

Answer: B

Explanation

Count BoW records the frequency of each word in the document, producing non-negative integers.

Question

The TF-IDF weight of term i in document j is:

- A. $TF_{ij} + IDF_i$
- B. $TF_{ij} \times IDF_i$
- C. TF_{ij} / IDF_i
- D. $TF_{ij} - IDF_i$

Answer: B

Explanation

TF-IDF = term frequency times inverse document frequency. It weights words that are frequent in a document but rare across the corpus.

Question

The inverse document frequency of term i across D documents is:

- A. $\ln(D/d_i)$
- B. $D \cdot d_i$
- C. $D + d_i$
- D. $D - d_i$

Answer: A

Explanation

$IDF = \ln(D/d)$, where d = number of documents containing term i . If term appears in all documents, $IDF = 0$.

Question

An n-gram in NLP is:

- A. A single word
- B. n contiguous words treated as a single unit
- C. A sentence
- D. A document

Answer: B

Explanation

An n-gram is n contiguous words treated as a unit. Uni-gram = 1 word, bi-gram = 2 words, etc.

Question

L2 normalization of a vector v divides each element by:

- A. $\sum v_i$
- B. $\sqrt{\sum v_i^2}$
- C. $\max(v_i)$
- D. $|v_1|$

Answer: B

Explanation

L2 norm = square root of sum of squares. Dividing by it produces a unit vector.

Question

The “naive” in Naive Bayes refers to:

- A. Assuming all words are independent
- B. Assuming all words are dependent
- C. Assuming equal class priors
- D. Assuming no training data

Answer: A

Explanation

Naive Bayes assumes features (words) are independent of each other, which simplifies computation but is rarely true.

Question

The Naive Bayes classification rule is:

- A. $\hat{c} = \arg \max_c P(c)$
- B. $\hat{c} = \arg \max_c P(w_1|c)P(w_2|c) \cdots P(w_N|c)P(c)$
- C. $\hat{c} = \arg \min_c P(c)$
- D. $\hat{c} = \sum_c P(c)$

Answer: B

Explanation

Naive Bayes predicts the class maximizing the product of conditional word probabilities and class prior.

Question

Laplace smoothing in Naive Bayes addresses:

- A. Computational overflow
- B. Zero probability problem
- C. Unknown words
- D. Class imbalance

Answer: B

Explanation

Laplace smoothing adds a small positive integer ($\lambda = 1$) to all counts, preventing zero probabilities when a word never appears in a class.

Question

Bayes' rule is:

- A. $P(c|d) = P(d|c)P(c) / P(d)$
- B. $P(c|d) = P(c) + P(d)$
- C. $P(c|d) = P(d|c) / P(c)$
- D. $P(c|d) = P(c) \cdot P(d)$

Answer: A

Explanation

Bayes: posterior = likelihood times prior divided by evidence. Used in Naive Bayes classification.

Question

The two architectures of Word2Vec are:

- A. CBoW and Skip-gram
- B. RNN and LSTM
- C. BERT and GPT
- D. CNN and RNN

Answer: A

Explanation

Word2Vec has two architectures: Continuous Bag of Words (CBoW) and Skip-gram.

Question

CBoW predicts target words from context, while Skip-gram predicts:

- A. Context words from target word
- B. Sentences from words
- C. Documents from words
- D. Words from sentences

Answer: A

Explanation

CBoW: predict target word from context words. Skip-gram: predict context words from target word.

Question

Cosine similarity between vectors P and Q is:

- A. Norm of P minus Q
- B. $P \cdot Q / (\|P\| \|Q\|)$
- C. $\|P\| + \|Q\|$
- D. P/Q

Answer: B

Explanation

Cosine similarity = dot product divided by product of magnitudes. Range is -1 to 1. Used in Word2Vec for word similarity.

Question

GloVe stands for:

- A. Global Vectors for Word Representation
- B. Global Learning of Vector Embeddings
- C. Generalized Local Vector Encoding
- D. Gradient Loss Vector Estimation

Answer: A

Explanation

GloVe = Global Vectors for Word Representation, developed at Stanford. It is an alternative to Word2Vec.

Question

In an RNN, the hidden state at time t depends on:

- A. Only the current input
- B. Only the previous hidden state
- C. Both the current input and the previous hidden state
- D. Neither

Answer: C

Explanation

RNN hidden state combines current input and previous hidden state, enabling memory of past information.

Question

An LSTM cell typically contains how many gates?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C

Explanation

LSTM has three gates: forget gate (discard), input gate (store), output gate (output). They regulate information flow.

Question

The forget gate in an LSTM determines:

- A. Which part of memory is discarded
- B. Which part of output is stored
- C. Which part of memory is output
- D. The learning rate

Answer: A

Explanation

Forget gate decides what to discard from memory. Input gate decides what to store. Output gate decides what to output.

Question

The 2017 paper introducing the transformer architecture is titled:

- A. Attention Is All You Need
- B. Deep Learning for AI
- C. Neural Networks and Learning
- D. Machine Learning Fundamentals

Answer: A

Explanation

Attention Is All You Need by Vaswani et al. (2017) introduced the transformer based on self-attention.

Question

BERT stands for:

- A. Bidirectional Encoder Representations from Transformers
- B. Bayesian Encoder for Recursive Tasks
- C. Binary Encoder Representation Technique
- D. Bidirectional Encoder for Random Text

Answer: A

Explanation

BERT = Bidirectional Encoder Representations from Transformers. Encoder-only model released by Google in 2018.

Question

GPT stands for:

- A. General Purpose Transformer
- B. Generative Pre-trained Transformer
- C. Gradient Pre-training Technique
- D. Global Parameter Training

Answer: B

Explanation

GPT = Generative Pre-trained Transformer. Decoder-only autoregressive model from OpenAI.

Question

BART stands for:

- A. Bidirectional and Auto-Regressive Transformer
- B. Bayesian Auto-Regressive Technique
- C. Binary Auto-Regressive Transformer
- D. Bidirectional Attention for Recognition Tasks

Answer: A

Explanation

BART = Bidirectional and Auto-Regressive Transformer. Combines encoder and decoder for understanding and generation.

Question

LSTM stands for:

- A. Long Short-Term Memory
- B. Linear Sequential Time Model
- C. Layered Stochastic Transformation Model
- D. Low-Signal Temporal Method

Answer: A

Explanation

LSTM = Long Short-Term Memory. Type of RNN that mitigates vanishing gradient via gating.

Question

The scaled dot-product attention score between query Q and key K is:

- A. $Q \cdot K$
- B. $Q \cdot K / \sqrt{d_k}$
- C. $(Q + K) / 2$
- D. $|Q - K|$

Answer: B

Explanation

Scaled dot-product attention divides by square root of key dimension to prevent softmax saturation.

Question

The softmax function is:

- A. $e^{x_i} / \sum_j e^{x_j}$
- B. $x_i / \sum_j x_j$
- C. e^{x_i}
- D. $\ln(x_i)$

Answer: A

Explanation

Softmax normalizes exponentials into a probability distribution. Used in attention and multiclass outputs.

Question

Context length (context window) of an LLM is:

- A. The number of parameters
- B. The maximum number of tokens the model can process in a single input/output
- C. The training time
- D. The vocabulary size

Answer: B

Explanation

Context length = maximum tokens processable in one sequence. GPT-4 has 32k; Gemini 2.5 Pro has 1048k.

Question

LLMs are stateless, meaning:

- A. They retain calculations between prompts
- B. They do not retain calculations or information between processing one prompt and the next
- C. They store all conversations
- D. They learn from every interaction

Answer: B

Explanation

Statelessness = no memory between prompts. Users must feed previous context if needed.

Question

Temperature in LLMs controls:

- A. The number of tokens generated
- B. The shape of the probability distribution (randomness/creativity)
- C. The context length
- D. The learning rate

Answer: B

Explanation

Temperature below 1 narrows distribution (more predictable). Temperature above 1 flattens distribution (more creative/random).

Question

Top-K sampling restricts the model to consider:

- A. All tokens
- B. The K most probable tokens at each step
- C. The K least probable tokens
- D. Random tokens

Answer: B

Explanation

Top-K sampling considers only the K most probable tokens. Low K sacrifices creativity; high K allows more variety.

Question

Top-P (nucleus) sampling selects:

- A. The K most probable tokens
- B. The smallest set of tokens whose cumulative probability is at least P
- C. All tokens
- D. Random tokens

Answer: B

Explanation

Top-P selects the smallest set of tokens with cumulative probability at least P. A low P sacrifices creativity.

Question

Chain-of-Thought (CoT) prompting:

- A. Asks for direct answers
- B. Prompts with intermediate reasoning steps
- C. Uses random prompts
- D. Removes context

Answer: B

Explanation

CoT prompting encourages step-by-step reasoning, improving performance on multi-step tasks like arithmetic.

Question

Zero-shot prompting involves:

- A. Providing one example
- B. Providing multiple examples
- C. Providing no examples, just the question
- D. Providing only context

Answer: C

Explanation

Zero-shot = no examples. One-shot = one example. Few-shot = multiple examples.

Question

Few-shot prompting involves:

- A. No examples
- B. One example
- C. Multiple examples
- D. Only context

Answer: C

Explanation

Few-shot = multiple relevant examples provided to guide the model's response.

Question

RAG stands for:

- A. Retrieval Augmented Generation
- B. Random Attention Gate
- C. Recurrent Adversarial Graph
- D. Regularized Attention Gradient

Answer: A

Explanation

RAG = Retrieval Augmented Generation. Augments LLM responses with retrieved documents for factual accuracy.

Question

RLHF stands for:

- A. Reinforcement Learning from Human Feedback
- B. Recursive Layered Hidden Features
- C. Random Loss with High Frequency
- D. Regularized Linear Hierarchical Framework

Answer: A

Explanation

RLHF = Reinforcement Learning from Human Feedback. Used in LLM training to align outputs with human preferences.

Question

Hallucination in GenAI refers to:

- A. Generating factually incorrect but plausible output
- B. Generating random noise
- C. Generating too many tokens
- D. Generating too few tokens

Answer: A

Explanation

Hallucination = factually incorrect, misleading, or fabricated output that appears credible.

Question

MMLU stands for:

- A. Massive Multitask Language Understanding
- B. Multi-Modal Language Understanding
- C. Machine Learning Utility
- D. Multi-Layer Language Understanding

Answer: A

Explanation

MMLU = Massive Multitask Language Understanding. Evaluates models across 57 subjects with multiple-choice questions.

Question

HELM stands for:

- A. Holistic Evaluation of Language Models
- B. Hierarchical Evaluation of Learning Models
- C. Hybrid Ensemble Learning Method
- D. High-Efficiency Language Model

Answer: A

Explanation

HELM = Holistic Evaluation of Language Models, a leaderboard from Stanford CRFM.

Question

In reinforcement learning, the agent is:

- A. The environment
- B. The learner or decision-maker
- C. The reward
- D. The state

Answer: B

Explanation

Agent = learner/decision-maker. In MAB, the agent is the gambler.

Question

In reinforcement learning, reward is:

- A. The state
- B. The action
- C. Feedback from the environment after taking an action
- D. The policy

Answer: C

Explanation

Reward = feedback (positive or negative) from the environment after an action. The agent aims to maximize cumulative reward.

Question

In reinforcement learning, a policy is:

- A. The reward function
- B. The agent's strategy for choosing actions given states
- C. The state space
- D. The action space

Answer: B

Explanation

Policy = strategy mapping states to actions. Goal: find optimal policy maximizing long-term reward.

Question

The value function $V(s)$ in reinforcement learning estimates:

- A. Immediate reward
- B. Total expected future reward from state s following a policy
- C. The number of states
- D. The action space size

Answer: B

Explanation

$V(s)$ = expected cumulative future reward starting from state s and following a policy.

Question

The action-value function $Q(s,a)$ estimates:

- A. Value of state s only
- B. Value of taking action a in state s
- C. Number of actions
- D. Reward only

Answer: B

Explanation

$Q(s,a)$ = expected cumulative future reward from taking action a in state s , then following a policy.

Question

The epsilon-greedy strategy balances:

- A. Only exploration
- B. Only exploitation
- C. Exploration (random action with probability epsilon) and exploitation (best action with probability 1 minus epsilon)
- D. Neither

Answer: C

Explanation

Epsilon-greedy: with probability epsilon explore randomly; with probability 1 minus epsilon exploit the best-known action.

Question

The greedy strategy in MAB:

- A. Always explores
- B. Always chooses the action with the best reward so far
- C. Chooses randomly
- D. Never chooses

Answer: B

Explanation

Greedy = pure exploitation. It always chooses the action with the highest estimated value, ignoring exploration.

Question

The random strategy in MAB:

- A. Exploits best action
- B. Randomly selects an action
- C. Never selects
- D. Uses gradient descent

Answer: B

Explanation

Random strategy = pure exploration. It randomly selects actions without using past knowledge.

Question

The Monte Carlo method in RL updates value estimates:

- A. After each step
- B. Only at the end of a complete episode
- C. Never
- D. Randomly

Answer: B

Explanation

Monte Carlo updates only at episode end using actual returns. High variance, unbiased.

Question

The Temporal Difference (TD) method in RL updates value estimates:

- A. Only at episode end
- B. After each step using bootstrapping
- C. Never
- D. Only at the beginning

Answer: B

Explanation

TD updates after each step by bootstrapping: using current estimate of next state's value. Lower variance than MC.

Question

Q-learning is a:

- A. Monte Carlo method
- B. Temporal Difference method
- C. Supervised learning method
- D. Unsupervised learning method

Answer: B

Explanation

Q-learning is a TD method that learns action-value function $Q(s,a)$ step-by-step.

Question

DQN stands for:

- A. Deep Q-Network
- B. Dynamic Quantum Network
- C. Dual Q-Network
- D. Distributed Q-Network

Answer: A

Explanation

DQN = Deep Q-Network. Combines TD method with neural networks to approximate $Q(s,a)$ for large state spaces.

Question

DQN consists of which three key components?

- A. Policy network, experience replay, target network
- B. Encoder, decoder, attention
- C. Input, hidden, output
- D. Actor, critic, reward

Answer: A

Explanation

DQN: policy network (Q-network), experience replay (memory buffer), target network (stable Q-values).

Question

Experience replay in DQN:

- A. Increases correlation between samples
- B. Breaks temporal correlation by random sampling of past experiences
- C. Removes outliers
- D. Increases learning rate

Answer: B

Explanation

Experience replay stores past experiences and samples mini-batches randomly, breaking temporal correlation.

Question

Policy-based RL methods:

- A. Learn value functions first
- B. Learn the policy directly
- C. Learn neither
- D. Only work with discrete actions

Answer: B

Explanation

Policy-based methods learn the policy directly (parameterized by θ), suitable for continuous action spaces.

Question

Actor-Critic methods combine:

- A. Value-based and policy-based approaches
- B. Supervised and unsupervised learning
- C. CNN and RNN
- D. Bagging and boosting

Answer: A

Explanation

Actor-Critic: actor (policy) plus critic (value function). Combines strengths of policy-based and value-based methods.

Question

The founders of utilitarianism are:

- A. Kant and Aristotle
- B. Bentham and Mill
- C. Rawls and Nozick
- D. Plato and Socrates

Answer: B

Explanation

Jeremy Bentham and John Stuart Mill developed classical utilitarianism in the 18th and 19th centuries.

Question

The categorical imperative is attributed to:

- A. Immanuel Kant
- B. John Stuart Mill
- C. Jeremy Bentham
- D. Aristotle

Answer: A

Explanation

Kant's categorical imperative: act only on maxims that could be universal laws; treat people as ends, not means.

Question

Virtue ethics is most closely associated with:

- A. Aristotle
- B. Kant
- C. Mill
- D. Bentham

Answer: A

Explanation

Virtue ethics has roots in Aristotle's *Nicomachean Ethics*, emphasizing virtuous character traits.

Question

Nonmaleficence refers to:

- A. Actively promoting welfare
- B. Avoiding harm and unjustified risk
- C. Fairness
- D. Respecting autonomy

Answer: B

Explanation

Nonmaleficence = do no harm. It forbids imposing unjustified risks.

Question

Beneficence refers to:

- A. Avoiding harm
- B. Actively promoting the welfare of others
- C. Respecting autonomy
- D. Ensuring fairness

Answer: B

Explanation

Beneficence = do good. It requires positive action to help others.

Question

Autonomy as an AI ethics principle requires:

- A. Respecting individual decision-making
- B. Ensuring fairness
- C. Avoiding harm
- D. Benefiting others

Answer: A

Explanation

Autonomy = respect for individual self-determination and consent.

Question

Justice as an AI ethics principle primarily concerns:

- A. Fairness, equality, and impartiality
- B. Avoiding harm
- C. Benefiting others
- D. Respecting autonomy

Answer: A

Explanation

Justice = fairness, equality, impartiality. Distributive justice concerns allocation; procedural justice concerns fair process.

Question

Consequentialism judges actions by:

- A. Adherence to rules
- B. Outcomes or consequences
- C. Character traits
- D. Legal compliance

Answer: B

Explanation

Consequentialism judges by outcomes. Utilitarianism is the most common form, maximizing overall welfare.

Question

Deontology judges actions by:

- A. Outcomes
- B. Adherence to duties and rules
- C. Character
- D. Profit

Answer: B

Explanation

Deontology judges by duties and rules irrespective of consequences. Kant is the primary proponent.

Question

Virtue ethics focuses on:

- A. Outcomes
- B. Rules
- C. Character traits and living a good life
- D. Legal compliance

Answer: C

Explanation

Virtue ethics asks what kind of person one should be, emphasizing virtues like wisdom, courage, and justice.

Question

Contextual integrity requires that:

- A. Data can be freely reused
- B. Data use must respect the norms of the context in which it was provided
- C. Data must be encrypted
- D. Data must be deleted

Answer: B

Explanation

Contextual integrity (Nissenbaum): data provided in one context should not be used in an incompatible context.

Question

Data minimisation requires:

- A. Collecting all available data
- B. Collecting only data necessary for the specified purpose
- C. Sharing data freely
- D. Retaining data indefinitely

Answer: B

Explanation

Data minimisation = collect only what is necessary for the specified purpose. Core GDPR principle.

Question

The right to be forgotten allows data subjects to:

- A. Access their data
- B. Request erasure of personal data under certain conditions
- C. Sell their data
- D. Share data without consent

Answer: B

Explanation

Right to erasure (GDPR Art. 17): allows deletion when data is no longer necessary, consent is withdrawn, etc.

Question

GDPR requires notification of a personal data breach to the supervisory authority within:

- A. 24 hours
- B. 48 hours
- C. 72 hours
- D. 1 week

Answer: C

Explanation

GDPR Article 33 requires breach notification within 72 hours of becoming aware of the breach.

Question

The GDPR right to erasure (right to be forgotten) is in:

- A. Article 6
- B. Article 17
- C. Article 33
- D. Article 5

Answer: B

Explanation

Article 17 = right to erasure. Article 6 = lawful basis; Article 33 = breach notification; Article 5 = principles.

Question

The NIST AI Risk Management Framework is organized around:

- A. Train, Test, Deploy, Monitor
- B. Govern, Map, Measure, Manage
- C. Design, Build, Audit, Retire
- D. Plan, Do, Check, Act

Answer: B

Explanation

NIST AI RMF = Govern (structures), Map (context), Measure (assessment), Manage (mitigation).

Question

Which of the following is prohibited under the EU AI Act?

- A. AI for medical diagnosis
- B. Social scoring based on social behavior
- C. AI for credit scoring
- D. AI for customer service

Answer: B

Explanation

Prohibited: social scoring, biometric categorization using sensitive characteristics, emotion recognition in workplace/education, untargeted facial scraping.

Question

Non-compliance with the EU AI Act can lead to fines up to:

- A. 10 million euros or 2 percent of global turnover
- B. 35 million euros or 7 percent of global turnover
- C. 5 million euros or 1 percent of global turnover
- D. 100 million euros or 10 percent of global turnover

Answer: B

Explanation

EU AI Act fines range from 7.5M euros (1.5 percent) to 35M euros (7 percent) of global turnover depending on offense and company size.

Question

SR 11-7, Guidance on Model Risk Management, was issued by:

- A. The Federal Reserve and the OCC
- B. The SEC
- C. The FDIC alone
- D. The CFTC

Answer: A

Explanation

SR 11-7 (2011) was issued jointly by the Federal Reserve and the Office of the Comptroller of the Currency.

Question

The Three Lines of Defense model was formalized by:

- A. The Institute of Internal Auditors (IIA)
- B. The Federal Reserve alone
- C. The Basel Committee alone
- D. The SEC alone

Answer: A

Explanation

The IIA formalized the Three Lines of Defense model, later updated to the Three Lines Model in 2020.

Question

A model inventory is:

- A. A list of all models in use, with owners, risk tiers, and validation status
- B. A list of data sources
- C. A list of employees
- D. A list of IT systems

Answer: A

Explanation

Model inventory = registry of all models. Includes owner, purpose, risk tier, validation status, and interdependencies.

Question

A model landscape is:

- A. A list of all models
- B. A map of interactions and interdependencies between models
- C. A training dataset
- D. A model architecture

Answer: B

Explanation

Model landscape = map of interactions and dependencies. It reveals cascade risk and critical nodes.

Question

Model risk appetite is:

- A. The set of models in production
- B. The level of model risk the organization is willing to accept
- C. The cost of modeling
- D. The number of models

Answer: B

Explanation

Model risk appetite = acceptable level of model risk, set by the board, guiding governance.

Question

The use test in model validation is:

- A. A quantitative accuracy metric
- B. A qualitative assessment of whether the model is used appropriately for its intended purpose
- C. A code review
- D. A speed test

Answer: B

Explanation

The use test assesses whether the model is used correctly and appropriately in practice. It is qualitative and people-focused.

Question

Outcomes analysis in model validation is used to:

- A. Speed up training
- B. Compare model predictions against realized outcomes to detect bias, degradation, or incorrect assumptions
- C. Reduce features
- D. Increase accuracy

Answer: B

Explanation

Outcomes analysis compares model predictions against actual outcomes to validate model quality in production.

Question

Initial validation serves which three key goals?

- A. Operational feasibility, proper documentation, user training
- B. Accuracy, speed, cost
- C. Data collection, feature selection, model choice
- D. Deployment, monitoring, retirement

Answer: A

Explanation

Initial validation ensures: (1) operational feasibility, (2) proper documentation and firm-wide standards, (3) user training.

Question

Effective challenge in model risk management refers to:

- A. A critical analysis by objective and informed parties to identify model assumptions and limitations
- B. A quantitative test
- C. A code review
- D. A data quality check

Answer: A

Explanation

SR 11-7's effective challenge: critical analysis by independent parties to identify assumptions, limitations, and weaknesses.

Question

According to SR 11-7, banks should conduct periodic review of each model at least:

- A. Monthly
- B. Quarterly
- C. Annually, or more frequently if warranted
- D. Every 3 years

Answer: C

Explanation

SR 11-7 recommends annual review at minimum, with more frequent reviews for higher-risk or AI/ML models.

Question

Compared to traditional models, AI/ML model validation requires:

- A. Less frequent monitoring
- B. More frequent monitoring and broader stakeholder involvement
- C. No validation
- D. Only code review

Answer: B

Explanation

AI/ML models require more frequent monitoring due to complexity, data drift, and broader stakeholder involvement (HR, compliance, operational risk).

Question

The model risk tier determines:

- A. The model's accuracy
- B. The depth and frequency of validation activities
- C. The number of features
- D. The training time

Answer: B

Explanation

Risk tier drives validation depth and frequency. Higher-risk models are validated more frequently.

Question

Model risk ranking typically includes which factors?

- A. Materiality, complexity, exposure
- B. Accuracy, speed, cost
- C. Data size, feature count, training time
- D. Color, shape, size

Answer: A

Explanation

Model risk ranking: materiality (impact), complexity (methodology), exposure (usage). These drive resource allocation.

Question

Data provenance refers to:

- A. Data storage format
- B. The origin, history, and legal right to use data
- C. Data encryption
- D. Data classification

Answer: B

Explanation

Data provenance = origin and legal rights to use data. Critical for alternative data and AI training sets.

Question

Data classification is the process of:

- A. Sorting data alphabetically
- B. Categorizing data by sensitivity and structure to determine handling and controls
- C. Encrypting data
- D. Compressing data

Answer: B

Explanation

Data classification assigns sensitivity labels (public, internal, confidential, restricted) and structure labels.

Question

Metadata is best defined as:

- A. Raw data
- B. Data about data
- C. Encrypted data
- D. Synthetic data

Answer: B

Explanation

Metadata = data about data. It documents origin, structure, definitions, and relationships.

Question

Role-Based Access Control (RBAC) grants access based on:

- A. Individual discretion
- B. User roles within the organization
- C. Random selection
- D. Data size

Answer: B

Explanation

RBAC = permissions based on organizational roles. ABAC = attribute-based; MAC = mandatory; DAC = discretionary.

Question

Attribute-Based Access Control (ABAC) grants access based on:

- A. User roles
- B. Attributes of user, resource, and environment
- C. Random selection
- D. Seniority

Answer: B

Explanation

ABAC uses attributes (user, resource, environment) to make access decisions. More flexible than RBAC.

Question

A data strategy should include:

- A. Vision, goals, priorities, and data governance policy
- B. Only data storage
- C. Only data encryption
- D. Only data deletion

Answer: A

Explanation

Data strategy includes vision, goals, priorities, authorization for non-public data, and approved data-governance policy.

Question

A data governance committee is typically responsible for:

- A. Operational data entry
- B. Approved data practices and policies, including collection, treatment, access, and monitoring
- C. Only data deletion
- D. Only data encryption

Answer: B

Explanation

Data governance committee provides oversight on all aspects of data governance (collection, reconciliations, treatment, access, monitoring).

Question

Synthetic data is used to:

- A. Replace real data entirely
- B. Augment available data and mitigate privacy risks
- C. Increase data storage
- D. Slow down training

Answer: B

Explanation

Synthetic data augments training data and mitigates risks when personally identifiable data may be present.

Question

AI-ready data is:

- A. Raw and unstructured
- B. Accurate, unbiased, well-structured, cleaned, and suitable for training
- C. Encrypted only
- D. Deleted data

Answer: B

Explanation

AI-ready data is accurate, unbiased, well-structured, cleaned, and suitable for training AI models.

Question

A data retention policy specifies:

- A. How long data is kept and when it is deleted
- B. Only data storage format
- C. Only data encryption
- D. Only data access

Answer: A

Explanation

Data retention policy defines how long data is kept and procedures for deletion, complying with regulations.

Question

Under GDPR, data subjects have the right to:

- A. Only access data
- B. Access, rectify, erase, restrict processing, data portability, and object
- C. Only delete data
- D. Only sell data

Answer: B

Explanation

GDPR rights: access, rectification, erasure, restriction, portability, objection, and rights related to automated decision-making.

Question

GDPR applies to organizations outside the EEA if they:

- A. Have any European customers
- B. Offer services to data subjects within the EEA or monitor their behavior
- C. Are publicly traded
- D. Have more than 100 employees

Answer: B

Explanation

GDPR has extraterritorial jurisdiction: applies to organizations outside EEA offering services to or monitoring EEA data subjects.

Question

CCPA stands for:

- A. California Consumer Privacy Act
- B. California Cyber Protection Act
- C. California Consumer Protection Act
- D. California Data Privacy Act

Answer: A

Explanation

CCPA = California Consumer Privacy Act (2018). Most comprehensive US state privacy law.

Question

Under CCPA, consumers have the right to:

- A. Know, access, delete, opt out of sale, and non-discrimination
- B. Only access data
- C. Only delete data
- D. Only sell data

Answer: A

Explanation

CCPA rights: know what data is collected, access, delete, opt out of sale, and non-discrimination.

Question

BIPA stands for:

- A. Biometric Information Privacy Act
- B. Business Information Protection Act
- C. Biometric Identification Protection Act
- D. Background Information Privacy Act

Answer: A

Explanation

BIPA = Biometric Information Privacy Act (Illinois, 2008). Regulates collection of biometric identifiers.

Question

Under BIPA, statutory damages are:

- A. 100 dollars per violation
- B. 1,000 dollars per violation (5,000 dollars if intentional/reckless)
- C. 10,000 dollars per violation
- D. No damages

Answer: B

Explanation

BIPA: 1,000 dollars per violation, or 5,000 dollars if intentional or reckless. Individuals can sue for damages.

Question

HIPAA stands for:

- A. Health Insurance Portability and Accountability Act
- B. Health Information Privacy and Accountability Act
- C. Health Insurance Privacy and Protection Act
- D. Health Information Portability Act

Answer: A

Explanation

HIPAA = Health Insurance Portability and Accountability Act (1996). Protects privacy and security of health information.

Question

GLBA stands for:

- A. Gramm-Leach-Bliley Act
- B. General Liability and Business Act
- C. Government-Led Banking Act
- D. Global Banking and Lending Act

Answer: A

Explanation

GLBA = Gramm-Leach-Bliley Act. Requires financial institutions to explain privacy practices and implement security safeguards.

Question

The FTC Safeguards Rule requires:

- A. Financial institutions to have measures to keep customer information secure
- B. Only data deletion
- C. Only data encryption
- D. Only data access

Answer: A

Explanation

FTC Safeguards Rule: financial institutions under FTC jurisdiction must have measures to keep customer information secure.

Question

The Privacy Act of 1974 governs:

- A. How federal agencies collect and use data about individuals
- B. Private sector data collection
- C. Only health data
- D. Only financial data

Answer: A

Explanation

Privacy Act of 1974: governs federal agencies' collection and use of personal data; prohibits disclosure without consent.

Question

The EU Digital Markets Act (DMA) targets:

- A. All companies
- B. Large online platforms with gatekeeper status
- C. Only banks
- D. Only healthcare companies

Answer: B

Explanation

DMA targets gatekeepers – large platforms with dominant positions. Bans self-preferencing, requires interoperability.

Question

The EU Digital Services Act (DSA) requires platforms to:

- A. Only remove illegal content
- B. Prevent illegal content, be transparent about moderation, address disinformation, protect against algorithmic bias
- C. Only label deepfakes
- D. Only allow opt-out

Answer: B

Explanation

DSA: prevent illegal content, transparent moderation, address disinformation, protect against algorithmic bias, allow opt-out.

Question

China's Generative AI Measures require service providers to:

- A. Only register with government
- B. Prohibit illegal content, prevent discriminatory content, not infringe on privacy rights
- C. Only label AI content
- D. Only obtain consent

Answer: B

Explanation

China's Generative AI Measures (2023): prohibit illegal content, prevent discrimination, protect privacy and personal information.

Question

PIPL stands for:

- A. Personal Information Protection Law
- B. Privacy and Information Protection Law
- C. Personal Identification Protection Law
- D. Public Information Privacy Law

Answer: A

Explanation

PIPL = Personal Information Protection Law (China). Mandates data minimization, user consent, and algorithmic transparency.

Question

The Colorado Privacy Act (CPA) gives consumers the right to:

- A. Only access data
- B. Know, access, delete, opt out of sale, and opt out of targeted advertising
- C. Only delete data
- D. Only sell data

Answer: B

Explanation

CPA (2021): know, access, delete, opt out of sale, and opt out of targeted advertising. Includes data minimization provisions.

Question

The Virginia Consumer Data Protection Act (CDPA) applies to businesses that:

- A. Have any Virginia customers
- B. Control or process personal data of at least 100,000 Virginia consumers
- C. Are publicly traded
- D. Have more than 10 employees

Answer: B

Explanation

CDPA applies to businesses controlling/processing data of 100,000 or more Virginia consumers, or 25,000 or more with more than 50 percent revenue from data sales.

Question

The Illinois Personal Information Protection Act (PIPA) requires:

- A. Only data encryption
- B. Reasonable security measures, breach notification within 24 hours, privacy notice, access, and opt-out
- C. Only data deletion
- D. Only data access

Answer: B

Explanation

PIPA: reasonable security, breach notification within 24 hours, privacy notice, access, correction, and opt-out of sale.

Question

The EU AI Act became effective on:

- A. August 1, 2024
- B. January 1, 2025
- C. March 13, 2024
- D. May 21, 2024

Answer: A

Explanation

EU AI Act effective August 1, 2024. Ban on unacceptable risk systems from February 2, 2025. Most provisions from August 2, 2026.

Question

The European AI Office is responsible for:

- A. Only research
- B. Compliance, implementation, and enforcement of the EU AI Act
- C. Only data protection
- D. Only cybersecurity

Answer: B

Explanation

European AI Office: first body in the world to enforce binding rules on AI. Responsible for compliance, implementation, enforcement.

Question

Providers of high-risk AI under the EU AI Act must:

- A. Only register
- B. Keep thorough records of datasets, methodologies, and oversight measures
- C. Only label content
- D. Only obtain consent

Answer: B

Explanation

High-risk AI providers must keep records of datasets, programming/training methodologies, and oversight measures. Impact assessment required.

Question

Under the EU AI Act, biometric categorization systems using sensitive characteristics are:

- A. Allowed with consent
- B. Prohibited
- C. Allowed for security
- D. Allowed for research

Answer: B

Explanation

Prohibited: biometric categorization using political, religious, philosophical beliefs, sexual orientation, race.

Question

Under the EU AI Act, emotion recognition in the workplace and educational institutions is:

- A. Allowed
- B. Prohibited
- C. Allowed with consent
- D. Allowed for safety

Answer: B

Explanation

Prohibited: emotion recognition in workplace and educational institutions under the EU AI Act.

Question

Under the EU AI Act, social scoring based on social behavior or personal characteristics is:

- A. Allowed
- B. Prohibited
- C. Allowed with consent
- D. Allowed for government

Answer: B

Explanation

Prohibited: social scoring based on social behavior or personal characteristics under the EU AI Act.

Question

Under the EU AI Act, AI systems that manipulate human behavior to circumvent free will are:

- A. Allowed
- B. Prohibited
- C. Allowed with consent
- D. Allowed for marketing

Answer: B

Explanation

Prohibited: AI systems that manipulate human behavior to circumvent free will, and AI exploiting vulnerabilities.

Question

Under the EU AI Act, untargeted scraping of facial images to create facial recognition databases is:

- A. Allowed
- B. Prohibited
- C. Allowed with consent
- D. Allowed for security

Answer: B

Explanation

Prohibited: untargeted scraping of facial images from internet or CCTV to create facial recognition databases.

Question

The EU AI Act requires companies to:

- A. Only register
- B. Notify people when interacting with chatbots, biometric systems, or emotion recognition; label deepfakes
- C. Only label content
- D. Only obtain consent

Answer: B

Explanation

AI Act transparency: notify when interacting with chatbots/biometric/emotion recognition; label deepfakes and AI-generated content.

Question

Under the EU AI Act, organizations deploying high-risk AI must:

- A. Only register
- B. Do an impact assessment on how AI affects people's rights
- C. Only label content
- D. Only obtain consent

Answer: B

Explanation

High-risk AI deployers (banks, insurance, essential services) must do an impact assessment on fundamental rights.

Question

EU AI Act fines range from:

- A. 1M to 10M euros
- B. 7.5M (1.5 percent) to 35M (7 percent) of global turnover
- C. 100M to 500M euros
- D. No fines

Answer: B

Explanation

EU AI Act fines: 7.5M or 1.5 percent to 35M or 7 percent of global turnover, depending on offense and company size.

Question

President Biden's Executive Order on Safe, Secure, and Trustworthy AI was issued on:

- A. October 30, 2023
- B. January 1, 2024
- C. May 12, 2021
- D. August 1, 2024

Answer: A

Explanation

EO on Safe, Secure, and Trustworthy Development and Use of AI issued October 30, 2023. Requires safety test results sharing with government.

Question

The AI Safety and Security Board was established by:

- A. The White House
- B. The Department of Homeland Security
- C. The SEC
- D. The FTC

Answer: B

Explanation

AI Safety and Security Board established by DHS on April 26, 2024, to advise on safe AI development in critical infrastructure.

Question

The NIST AI RMF's four functions are:

- A. Govern, Map, Measure, Manage
- B. Plan, Do, Check, Act
- C. Train, Test, Deploy, Monitor
- D. Design, Build, Audit, Retire

Answer: A

Explanation

NIST AI RMF: Govern (structures), Map (context), Measure (assessment), Manage (mitigation). Voluntary framework.

Question

The NIST AI Risk Management Framework is:

- A. Mandatory
- B. Voluntary and non-prescriptive
- C. Only for government
- D. Only for banks

Answer: B

Explanation

NIST AI RMF is voluntary, non-prescriptive, flexible, and customizable based on specific use cases and risk profiles.

Question

SR 11-7 defines a model as:

- A. Any spreadsheet
- B. A quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories to process input data into quantitative estimates
- C. Only AI models
- D. Only machine learning models

Answer: B

Explanation

SR 11-7: model = quantitative method/system/approach applying theories and assumptions to process input data into quantitative estimates.

Question

Model risk is defined as:

- A. The risk of data loss
- B. The risk of adverse consequences from decisions based on incorrect or misused model outputs
- C. The risk of cyber attack
- D. The risk of model theft

Answer: B

Explanation

Model risk = risk of adverse consequences from decisions based on incorrect or misused model outputs and reports.

Question

A comprehensive model inventory should include:

- A. Only model names
- B. Owner, purpose, methodology, risk tier, validation status, interdependencies
- C. Only model code
- D. Only model data

Answer: B

Explanation

Model inventory includes: owner, purpose, methodology, risk classification, validation status, interdependencies, and more.

Question

The model owner is responsible for:

- A. Only model development
- B. Proper development, validation, approval, remediation, implementation, use, monitoring, and change management
- C. Only model use
- D. Only model documentation

Answer: B

Explanation

Model owner oversees the model lifecycle: development, validation, approval, findings remediation, implementation, use, monitoring, and change management.

Question

Model developers are responsible for:

- A. Only coding
- B. Creating models, documenting decisions, and responding to validation feedback
- C. Only testing
- D. Only deployment

Answer: B

Explanation

Model developers create models, document modeling decisions, and act on validation feedback to improve models.

Question

Model validators are responsible for:

- A. Developing models
- B. Providing independent and objective evaluation of models, assumptions, and methodologies
- C. Only coding
- D. Only data collection

Answer: B

Explanation

Model validators provide independent and objective evaluation, reviewing assumptions, methodologies, and communicating findings to owners/developers.

Question

Internal and external auditors in model risk management:

- A. Are the primary model validators
- B. Provide independent checks on the validation process and outputs
- C. Develop models
- D. Own models

Answer: B

Explanation

Auditors provide meta-validation – independent checks on the validation process and outputs, not primary validation.

Question

Regulatory authorities may review:

- A. Only financial statements
- B. Risk models' validation processes and outcomes as part of supervision
- C. Only IT systems
- D. Only data

Answer: B

Explanation

Regulators review risk model validation processes and outcomes, including internal audits and compliance with regulatory requirements.

Question

A model risk committee is used to:

- A. Only develop models
- B. Ensure comparable information levels among participants and shared responsibility for difficult decisions
- C. Only approve models
- D. Only validate models

Answer: B

Explanation

Model risk committee ensures comparable information and shared responsibility for difficult decisions.

Question

The first line of defense in model risk management is:

- A. Independent risk management
- B. Business and model owners/developers
- C. Internal audit
- D. Regulators

Answer: B

Explanation

First line: business and model owners/developers. They own the model and its performance, monitor metrics, and react to breaches.

Question

The second line of defense in model risk management is:

- A. Business units
- B. Independent risk management/model validation
- C. Internal audit
- D. Regulators

Answer: B

Explanation

Second line: independent risk management/model validation. Sets policy, conducts validations, challenges first line, maintains inventory.

Question

The third line of defense in model risk management is:

- A. Business units
- B. Independent risk management
- C. Internal audit
- D. Regulators

Answer: C

Explanation

Third line: internal audit. Provides independent assurance to the board that the model risk management framework is well-designed and operating effectively.

Question

For black-box AI/ML models, validation should focus on:

- A. Only code review
- B. Explainability, benchmarking, assessment on different datasets, and back testing
- C. Only data quality
- D. Only model accuracy

Answer: B

Explanation

Black-box validation: focus on explainability, benchmarking against classical models, assessment on different datasets, and back testing.

Question

Benchmarking in AI/ML model validation involves:

- A. Only comparing accuracy
- B. Comparing model results to classical models to understand deviations
- C. Only data quality checks
- D. Only code review

Answer: B

Explanation

Benchmarking: comparison of AI/ML model results to classical models to understand reasons for deviation.

Question

SR 11-7's advice on analyzing in-sample fit and model performance in holdout samples is:

- A. Not applicable to AI/ML
- B. One of the main applicable tests for AI/ML models
- C. Only for traditional models
- D. Only for vendor models

Answer: B

Explanation

SR 11-7's holdout sample analysis is one of the main applicable tests for AI/ML models.

Question

For AI/ML models, common back testing methods like MSE may become ineffective because:

- A. MSE is always effective
- B. There may no longer be a straightforward relation between inputs and outputs
- C. MSE is too complex
- D. MSE is not available

Answer: B

Explanation

For AI/ML, MSE may be ineffective due to non-straightforward input-output relations. K-fold cross-validation may be more suitable.

Question

For AI/ML models, which technique might be more suitable for back testing than MSE?

- A. K-fold cross-validation
- B. Only R-squared
- C. Only p-values
- D. Only t-statistics

Answer: A

Explanation

K-fold cross-validation is more suitable for AI/ML back testing than traditional MSE due to non-linear input-output relations.

Question

The reporting component in AI/ML validation verifies that model outputs are:

- A. Only accurate
- B. Accurate, complete, informative, and contain appropriate indicators of model performance and limitations
- C. Only fast
- D. Only cheap

Answer: B

Explanation

Reporting component: verify outputs are accurate, complete, informative, and contain indicators of performance and limitations.

Question

Assessment of specific cases in AI/ML validation includes:

- A. Only extreme input values
- B. Stress testing with extreme values and analysis of borderline decisions
- C. Only average cases
- D. Only random cases

Answer: B

Explanation

Specific case assessment: stress testing with extreme input values and analysis of decisions made for borderline cases.

Question

For AI/ML models, explainability is key because:

- A. Models are always transparent
- B. Models are often less than transparent and may be black-box
- C. Models are simple
- D. Models are always correct

Answer: B

Explanation

AI/ML models may be less transparent (black-box). Explainability is key; outcomes must be rigorously tested and compared with traditional models.

Question

Which of the following is a challenge in validating GenAI models?

- A. Deterministic output
- B. Open-ended output and hallucination
- C. Simple performance metrics
- D. No data drift

Answer: B

Explanation

GenAI validation challenges: open-ended output, complex performance metrics, data drift, hallucination, bias, multi-modal issues, and interpretability.

Question

Hallucination in GenAI models refers to:

- A. Generating factually incorrect but convincing output
- B. Generating random noise
- C. Generating too many tokens
- D. Generating too few tokens

Answer: A

Explanation

Hallucination: factually incorrect, misleading, or fabricated output that appears credible and convincing.

Question

Data drift in GenAI models refers to:

- A. Changes in model architecture
- B. Changes in underlying environment making model responses obsolete
- C. Changes in training data only
- D. Changes in learning rate

Answer: B

Explanation

Data drift: with passage of time, model responses may become irrelevant due to changes in underlying environment, making model obsolete.

Question

AI models can amplify bias inherent in data because:

- A. They are unbiased
- B. They learn and reproduce patterns, including discriminatory ones
- C. They ignore data
- D. They only use clean data

Answer: B

Explanation

AI models can amplify bias in data. Proper attention must ensure model responses are not discriminatory or biased.

Question

Multi-modal issues in GenAI validation arise from:

- A. Only text data
- B. Processing images, video, and audio in addition to text
- C. Only structured data
- D. Only numerical data

Answer: B

Explanation

Multi-modal issues: with ability to process images, video, and audio in addition to text, validating GenAI models is more challenging.

Question

For GenAI models, human review should be:

- A. Avoided
- B. Part of the validation process
- C. Only for deployment
- D. Only for training

Answer: B

Explanation

Human review must be part of the validation process for GenAI models due to their unique challenges.

Question

Particular attention should be paid to validating:

- A. Only in-house models
- B. Open-source and third-party AI models offered as models as a service
- C. Only vendor models
- D. Only traditional models

Answer: B

Explanation

Attention to open-source and third-party AI models offered as models as a service, which are increasingly common.

Question

GenAI models require significant computational resources for:

- A. Only training
- B. Training, validation, and recalibration
- C. Only inference
- D. Only storage

Answer: B

Explanation

GenAI models require significant computer resources to train, validate, and recalibrate.

Question

GenAI model results are difficult to:

- A. Generate
- B. Explain or interpret
- C. Store
- D. Delete

Answer: B

Explanation

GenAI model results are difficult to explain or interpret, making validation challenging.

Question

The rapidly changing technological development in GenAI requires:

- A. Less frequent review
- B. More frequent review and recalibration of models
- C. No review
- D. Only initial validation

Answer: B

Explanation

GenAI is a rapidly changing area requiring more frequent review and recalibration of models.

Question

GenAI performance metrics are:

- A. Simpler than traditional ML
- B. More complex and varied than traditional ML
- C. The same as traditional ML
- D. Nonexistent

Answer: B

Explanation

GenAI performance metrics are more complex and varied than traditional ML due to open-ended, unstructured output.

Question

Higher temperature in GenAI models leads to:

- A. More predictable output
- B. More creative/random output
- C. No change
- D. Faster output

Answer: B

Explanation

Higher temperature widens probability distribution, leading to more creative/random output. Lower temperature = more predictable.

Question

Lower Top-K in GenAI models leads to:

- A. More creativity
- B. Less creativity (more predictable)
- C. No change
- D. More tokens

Answer: B

Explanation

Lower Top-K restricts to fewer most probable tokens, reducing creativity. Higher Top-K allows more variety.

Question

Lower Top-P in GenAI models leads to:

- A. More creativity
- B. Less creativity (more predictable)
- C. No change
- D. More tokens

Answer: B

Explanation

Lower Top-P restricts to smaller set of tokens, reducing creativity. Higher Top-P allows more variety.

Question

In OpenAI models, temperature ranges from:

- A. 0 to 1
- B. 0 to 2
- C. -1 to 1
- D. 1 to 10

Answer: B

Explanation

OpenAI temperature ranges from 0 to 2. Below 1 = more predictable; above 1 = more random.

Question

When temperature = 1 in an LLM:

- A. Distribution is narrowed
- B. Distribution is widened
- C. Probability distribution is not altered
- D. Model stops working

Answer: C

Explanation

Temperature = 1 means probability distribution is not altered. Below 1 = narrowed; above 1 = widened/flattened.

Question

Tokenization in GenAI refers to:

- A. Removing stop words
- B. Separating text into smaller units (tokens)
- C. Converting words to vectors
- D. Replacing words with synonyms

Answer: B

Explanation

Tokenization: separating text into smaller units (tokens) for processing by LLMs.

Question

Word embeddings in GenAI are:

- A. Sparse vectors
- B. Dense vector representations of words
- C. One-hot vectors
- D. Binary vectors

Answer: B

Explanation

Word embeddings are dense vector representations capturing semantic meaning. Word2Vec and GloVe are examples.

Question

The attention mechanism in transformers:

- A. Ignores word relationships
- B. Computes similarity between query and key vectors to weight values
- C. Only processes one word at a time
- D. Uses no weights

Answer: B

Explanation

Attention: computes similarity between query and key vectors, uses softmax weights to combine value vectors.

Question

Multi-head attention in transformers:

- A. Uses one attention head
- B. Uses multiple attention heads to capture different relationships
- C. Ignores relationships
- D. Uses no weights

Answer: B

Explanation

Multi-head attention uses several attention heads, each focusing on different aspects of the sequence.

Question

Positional embeddings in transformers:

- A. Are not used
- B. Provide information about the position of words in a sequence
- C. Only encode word meaning
- D. Only encode syntax

Answer: B

Explanation

Positional embeddings provide information about word positions, using sine/cosine functions or learned parameters.